DEPI IBM Data Science

Employee Attrition Prediction & Analysis Project

Final Presentation — Model, Deployment & Business Impact

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Agenda

01 **Business Problem & Objectives**

02 **Dataset & Exploratory Analysis**

03 **Statistical Validation & Feature Selection**

04 **Model Development & Optimization**

05 **Deployment Architecture & Live Demo**

06 **Business Impact & Next Steps**

The Business Problem

Employee attrition carries direct and hidden costs — recruitment, onboarding, lost productivity, and knowledge drain. Identifying at-risk employees early lets HR intervene before it's too late.

Our Objectives

- Identify the strongest drivers of attrition using statistics and feature importance
- Benchmark multiple ML models for the best accuracy / recall trade-off
- Deploy the model as a live, usable tool — not just a notebook
- Add monitoring so prediction quality can be tracked over time

Why it matters

16.1%

of employees in our data left the company

~ EGP 1,000s

avg. income gap between leavers and stayers

Top signal

Overtime is the #1 recurring driver across every test

Dataset Overview

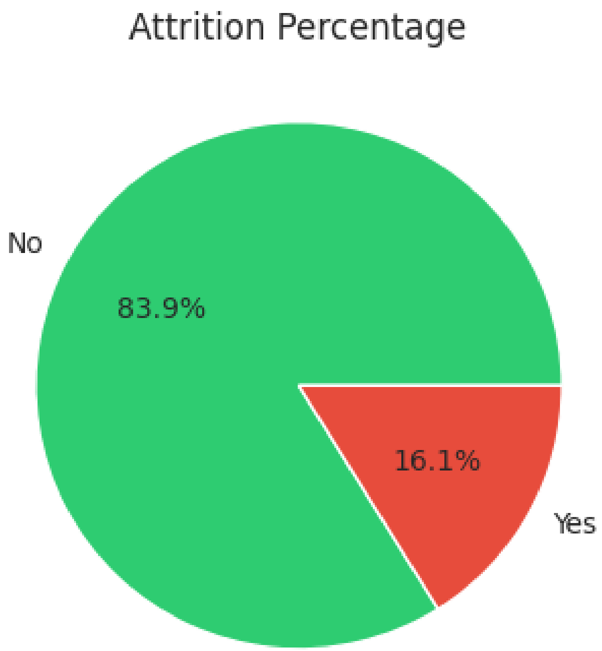
IBM HR Analytics Employee Attrition & Performance dataset

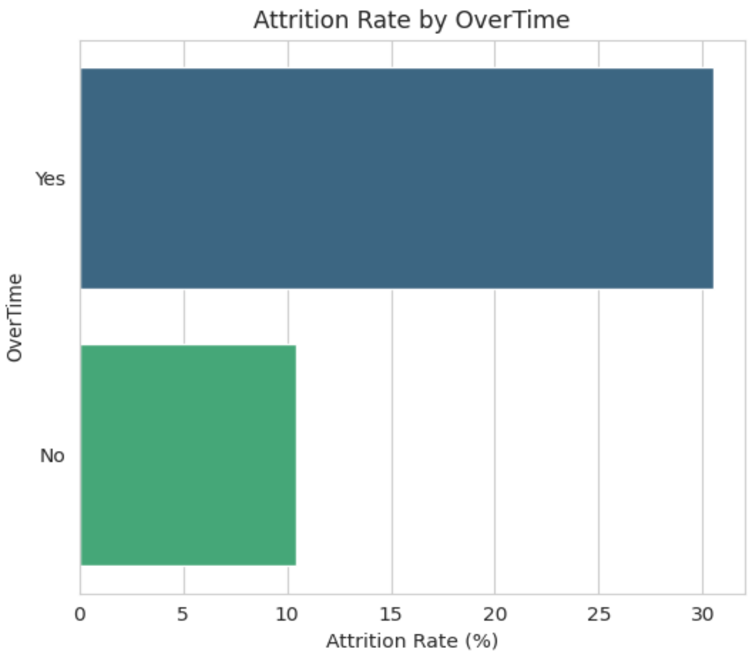
1,470	35	0	83.9% / 16.1%
Employee records	Features (numeric + categorical)	Missing values or duplicates	Stayed vs. Left (imbalanced)

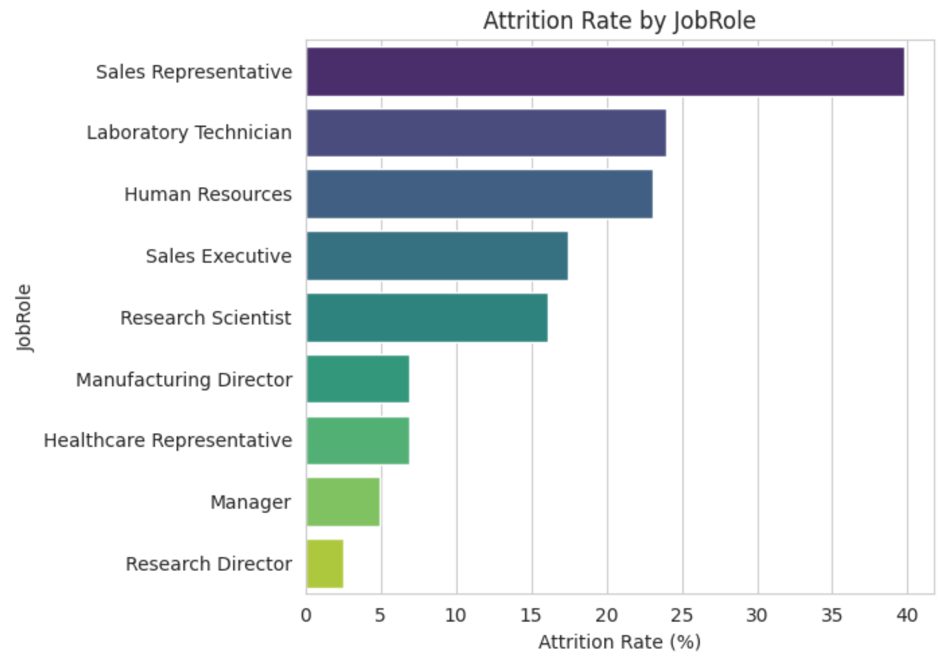
Because the data had no missing values or duplicates, preprocessing focused on encoding categorical variables, engineering new features (SalaryExperienceRatio, TenureGroup), and correcting class imbalance with SMOTE — rather than data cleaning.

Dataset Overview

IBM HR Analytics Employee Attrition & Performance dataset







Key EDA Insights

- Most employees are 30–40 years old; Monthly Income is right-skewed
- Employees who work overtime leave at a much higher rate
- Attrition varies meaningfully by department and job role
- Most employees rate work-life balance as 3 (“Better”) on a 1–4 scale
- Outliers concentrated in Monthly Income & Total Working Years

Correlation Highlights

- YearsAtCompany, YearsInCurrentRole & YearsWithCurrManager are strongly correlated
- MonthlyIncome correlates with JobLevel and TotalWorkingYears
- Tenure and role stability move together — employees who stay longer keep the same role and manager

Statistical Validation

Hypothesis tests confirm which features truly drive attrition

Test	Comparison	Result	Conclusion
t-test	Monthly Income: Left vs Stayed	$t=-7.48$, $p<0.001$	Leavers earn significantly less
Chi-squared	Job Role vs Attrition	$\chi^2=86.19$, $p<0.001$	Attrition differs by job role
Chi-squared	OverTime vs Attrition	$\chi^2=87.56$, $p<0.001$	Overtime strongly linked to leaving
ANOVA	Income across Job Roles	$F=810.21$, $p<0.001$	Income differs sharply by role
ANOVA	Income across Work-Life Balance	$F=0.61$, $p=0.607$	Not significant

Bottom line: Salary, overtime, and job role are statistically validated drivers of attrition — directly shaping our feature selection and HR recommendations.

Feature Engineering & Selection

Engineered Features

- SalaryExperienceRatio — balances pay against work experience
- TenureGroup — bins YearsAtCompany into New / Junior / Mid / Senior career stages

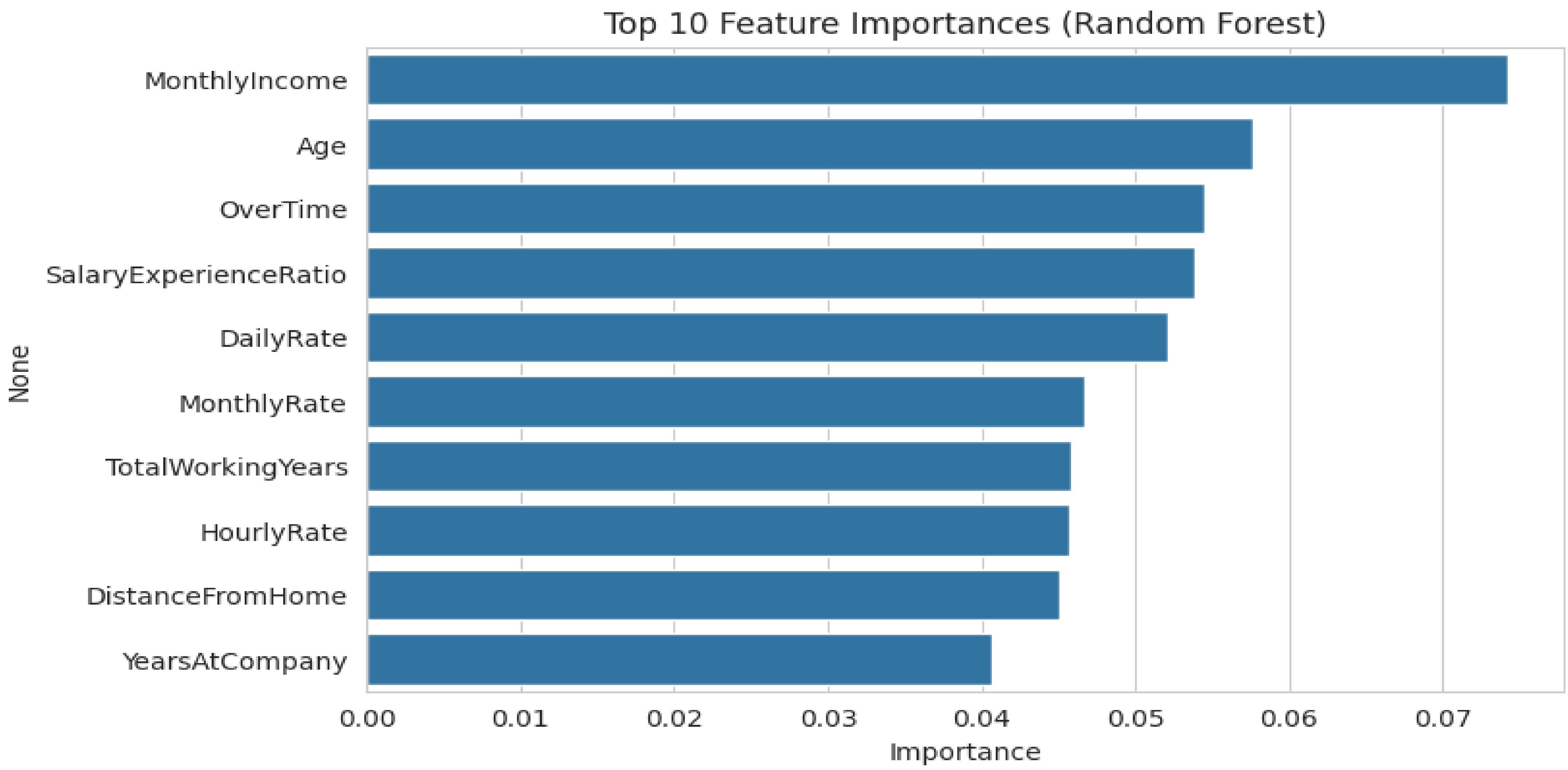
Selection Techniques

- RFE (Logistic Regression): 10 features selected, incl. SalaryExperienceRatio
- Random Forest importance: MonthlyIncome, Age, OverTime lead the ranking
- Final set = union of both techniques → Employee_Attrition_Selected_Features.csv

Consistently Top-Ranked

- MonthlyIncome
- OverTime
- SalaryExperienceRatio
- Age
- TotalWorkingYears

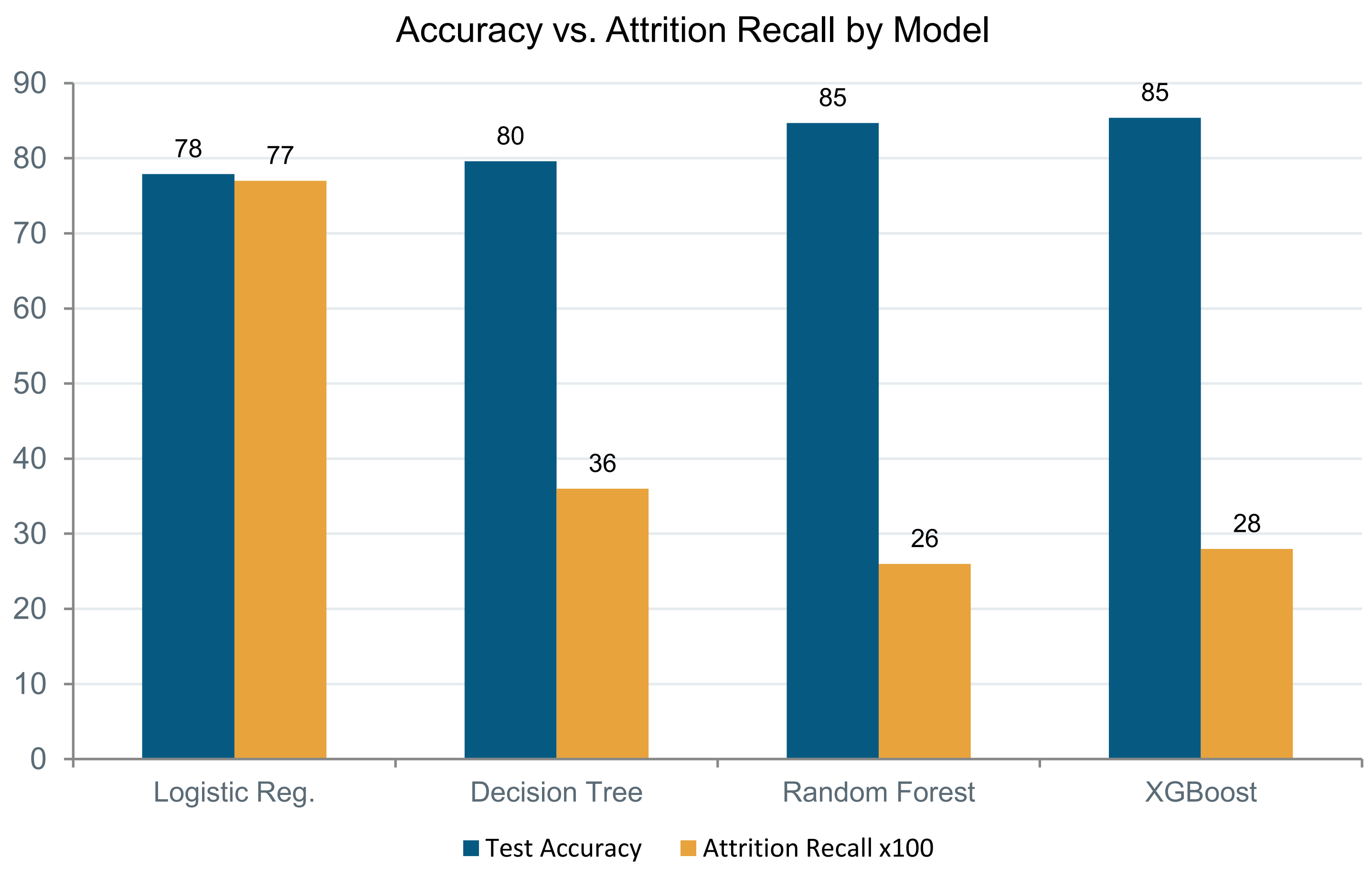
graphs



Top 10 Feature Importances (Random Forest)
MonthlyIncome
Age
OverTime
SalaryExperienceRatio
DailyRate
MonthlyRate
TotalWorkingYears
HourlyRate
DistanceFromHome
YearsAtCompany
None
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Importance

Model Development

4 classifiers, SMOTE-balanced training data, 5-fold stratified CV



Why Accuracy Isn't Enough

- Random Forest & XGBoost score highest on accuracy...
- ...but Logistic Regression catches far more real leavers (77% recall)
- For attrition, missing a true leaver is costlier than a false alarm

Optimizing the Final Model

GridSearchCV Tuning (XGBoost)

Parameter	Best Value
learning_rate	0.01
max_depth	4
n_estimators	400
Best CV F1	0.8831

27 combinations × 5-fold CV = 135 fits tested

Decision Threshold Tuning

Default 0.5 cutoff under-catches leavers. Testing 0.20–0.50 found the best F1 (0.50) at a threshold of 0.25 — adopted for production.

Final Tuned Model — Test Set

82.99%

Test Accuracy

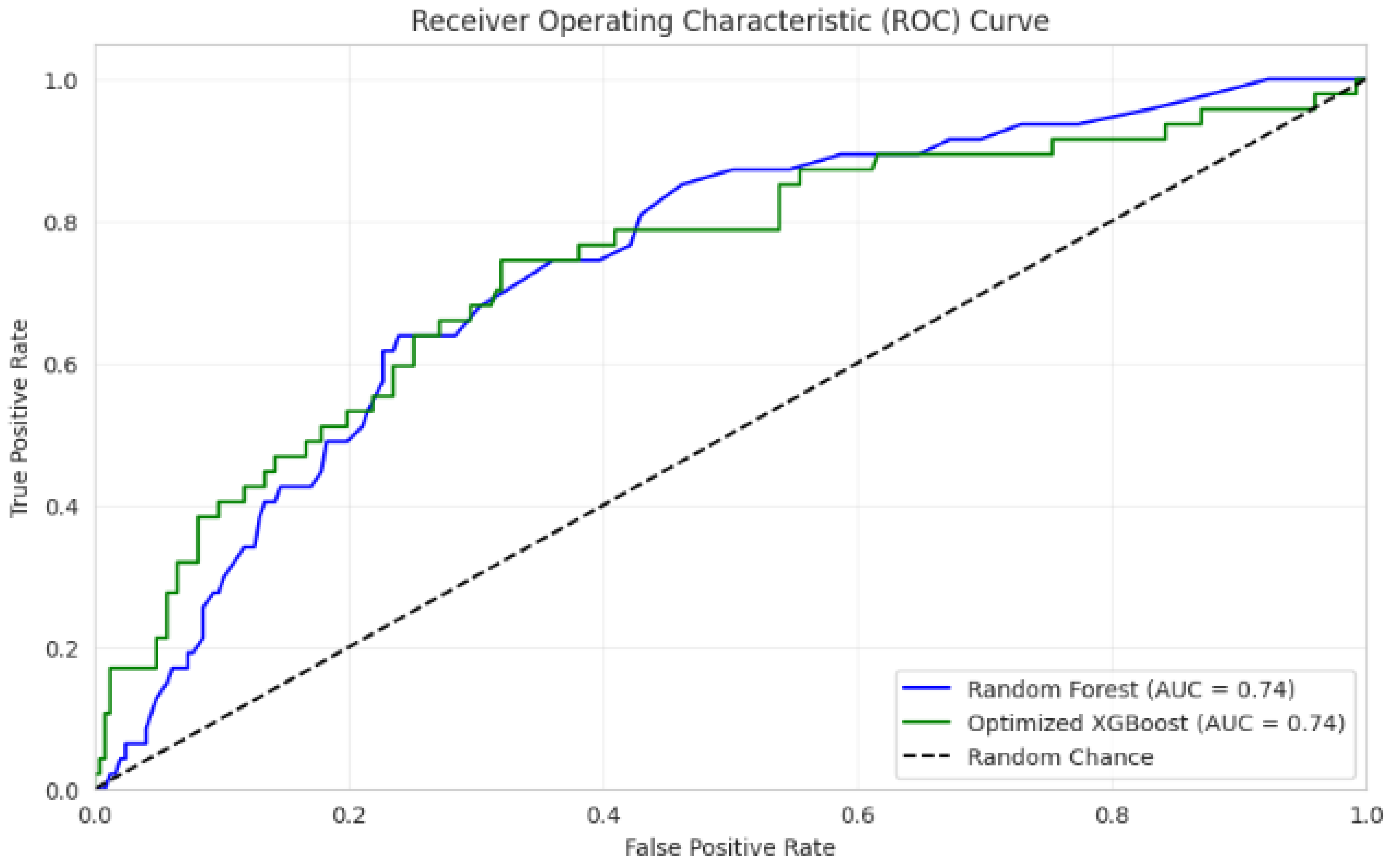
0.79

ROC-AUC

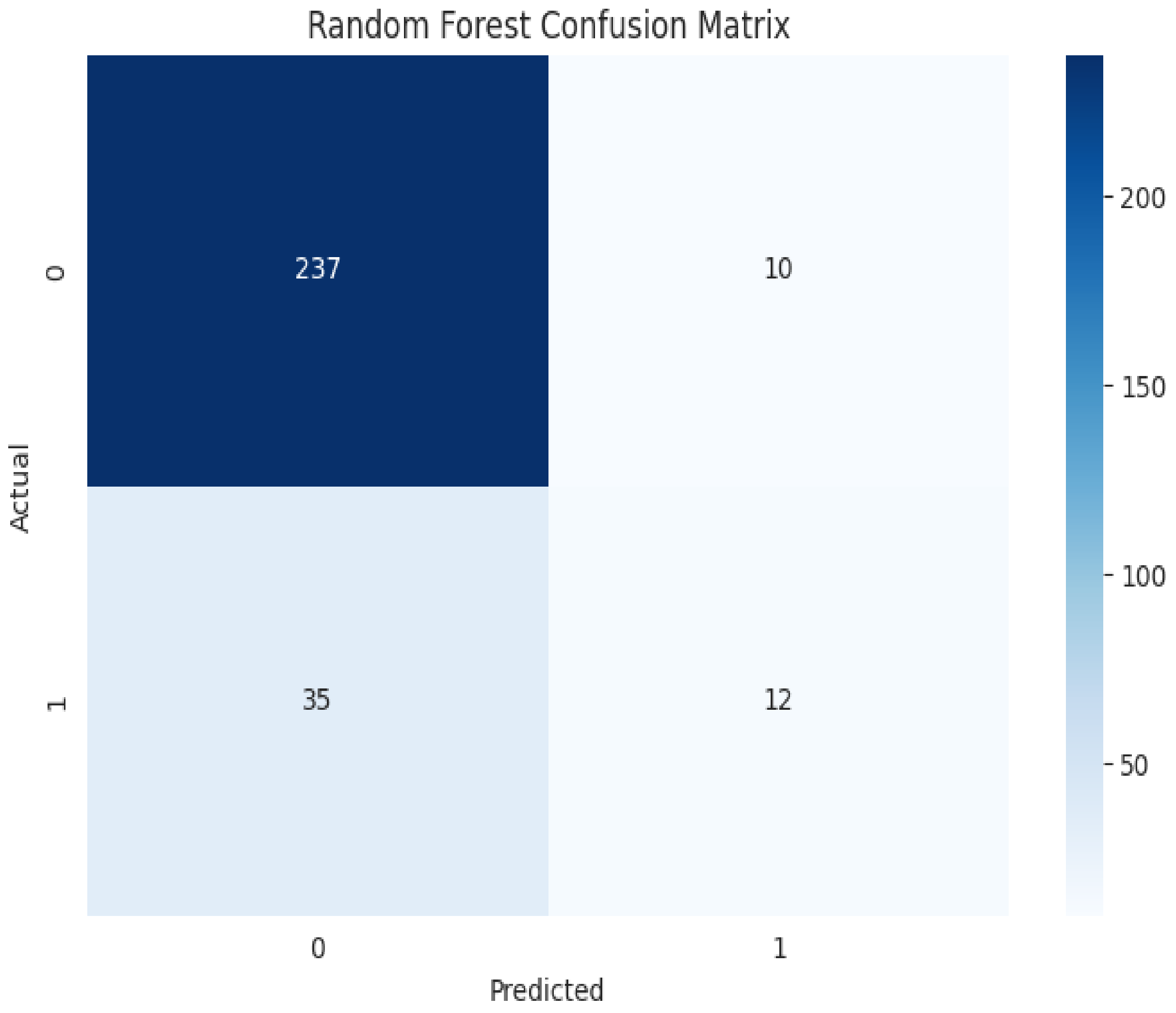
0.45 / 0.30

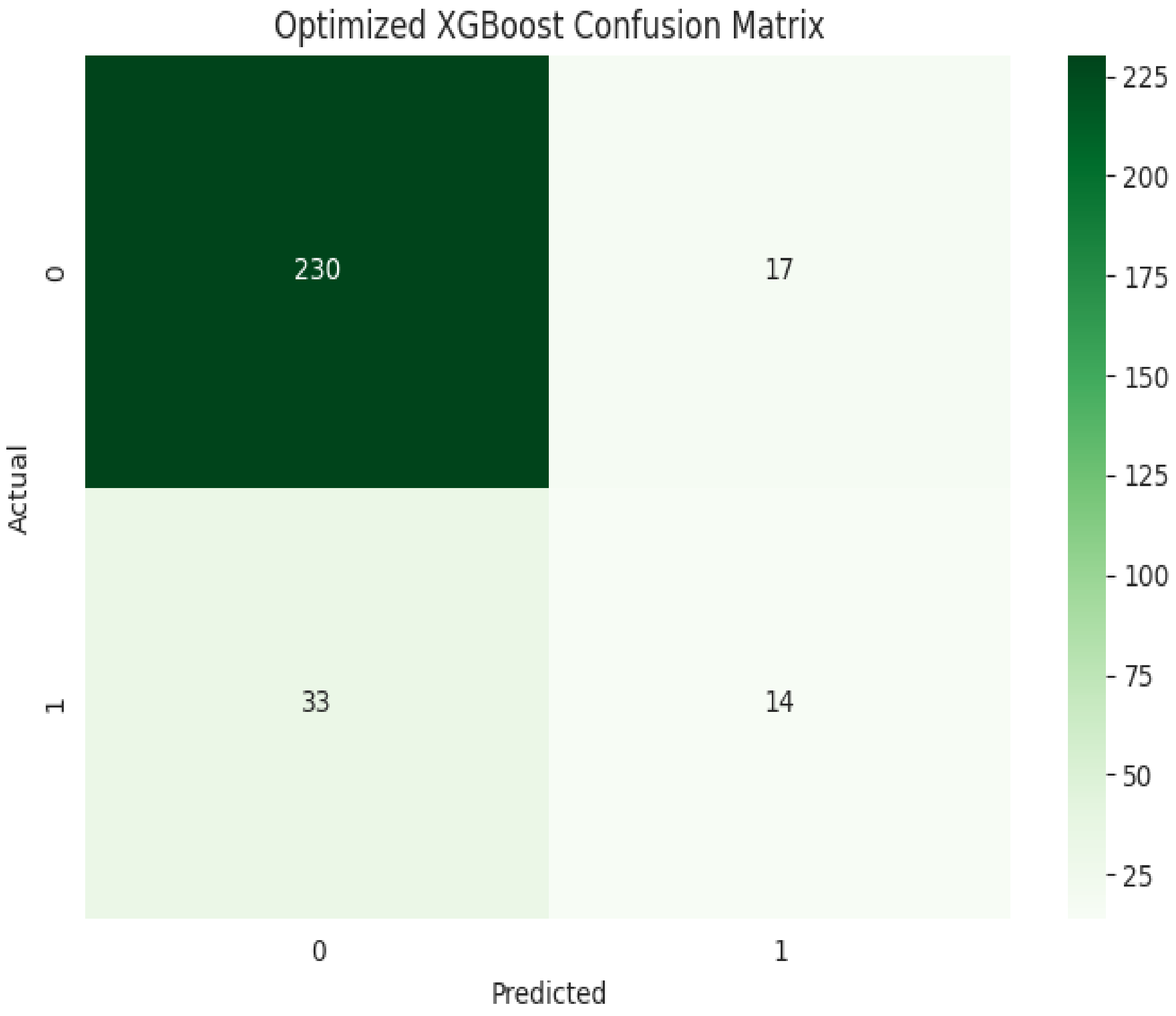
Precision / Recall (Attrition = Yes)

graphs


Receiver Operating Characteristic (ROC) Curve
1.0
0.8
0.6
0.4
0.2
0.0
True Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
False Positive Rate
Random Forest (AUC = 0.74)
Optimized XGBoost (AUC = 0.74)
Random Chance

graphs


Random Forest Confusion Matrix
237
10
35
12
Actual
0
1
Predicted
0
1
200
150
100
50


Optimized XGBoost Confusion Matrix
230
17
33
14
Actual
0
1
Predicted
0
1
225
200
175
150
125
100
75
50
25

Deployment Architecture

Trained Model

XGBoost + scaler +
feature artifacts (.pkl)

FastAPI Backend

/predict endpoint
hosted on Railway

Streamlit Dashboard

HR-facing UI
Streamlit Community Cloud

- Railway resolved earlier module-path & scikit-learn version-pinning issues from initial setup
- An initial Vercel deployment was abandoned due to serverless package size limits
- Every prediction is logged (input, prediction, probability, timestamp) for monitoring

Live Demo

HR Attrition Risk Dashboard — Streamlit

🔗

[https://employee-attrition-prediction-and-analysis-project.streamlit.app/]

Enter an employee's Age, Monthly Income, OverTime status, Total Working Years, and Job Satisfaction to get an instant attrition risk score.

Business Impact & HR Recommendations

- Review overtime policy — the strongest, most consistent driver of attrition
- Address pay gaps in high-turnover roles — leavers earn significantly less on average
- Watch tenure milestones — role & manager stability compound over time
- Use the dashboard proactively — score teams before resignations happen, not after
- Recall over raw accuracy — threshold tuned to catch more true leavers

Future Improvements

1

Richer Features

Add engagement & satisfaction survey scores

2

New Algorithms

Try neural networks, LightGBM, CatBoost

3

Automated Retraining

Use monitoring logs to trigger drift-based refreshes

4

Scalable Deployment

Containerize with Docker; expand dashboard inputs

5

Explainability

Add SHAP values so HR sees *why* an employee is flagged

رواد مصر الرقمية

وزارة الاتصالات وتكنولوجيا المعلومات

Thank You

Questions & Discussion